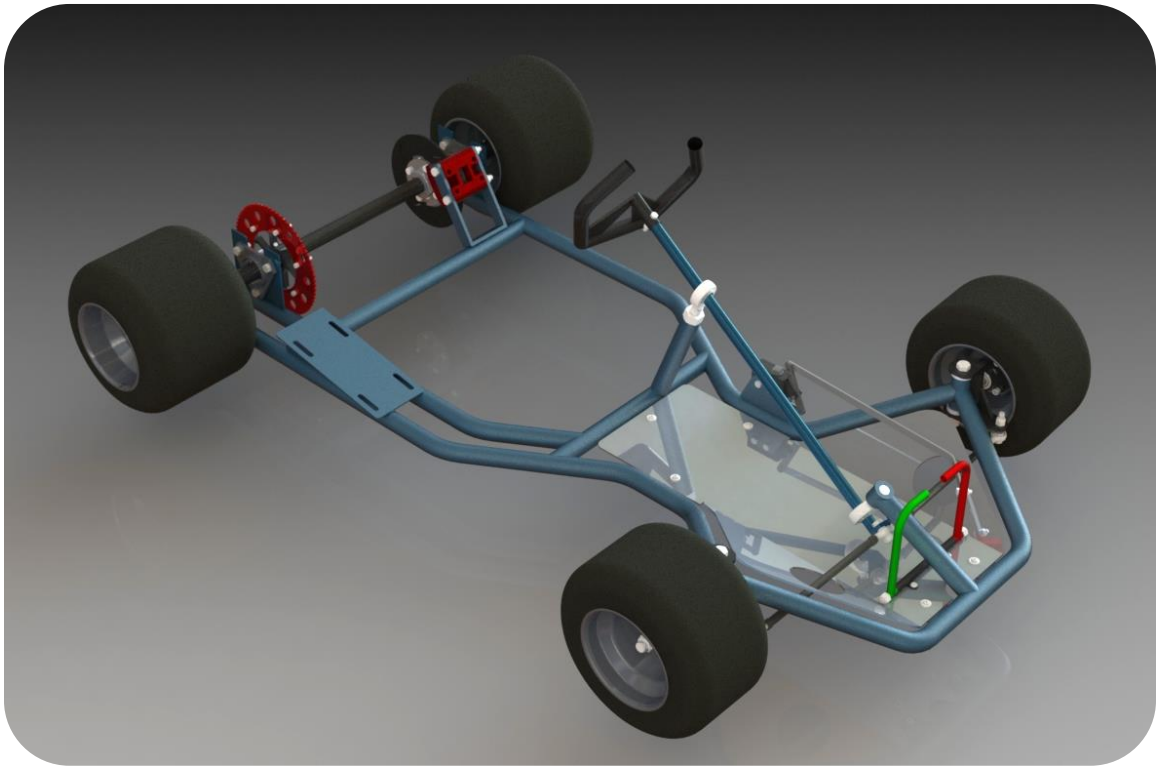
GC 498 Independent Study Report

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Proposal

Instructor/Sponsor: Dr. Nolan Fahrer

Student: Logan Griffin

Term: 2021 Spring

Units: 3.00

Meetings: Upon request, no set schedule

Purpose:

The purpose of this independent study is to apply knowledge gained from previous Graphic Communications courses and improve modeling skills through the modeling of an existing go-kart design and exercises from a Solidworks textbook.

Personal Motivation:

In high school, I was able to take Project Lead the Way (PLTW) classes and through those classes I was introduced to 3D modeling. I found it very interesting and wanted to take courses on it in college. Therefore, I decided to pursue a minor in Graphic Communications.

I have taken GC 120, GC 250, GC 350, and I wanted to take GC 450 to complete the coursework required to receive the minor. Since GC 450 will not be offered for the 2021 spring semester, this independent study will allow me to study the topics that are covered in that class.

In addition, working through textbook exercises will allow me to expand my knowledge of the features within Solidworks. Walking through exercises step by step will hep me understand features in the program I do not have any experience with.

Completing this independent study will result in the application of prior Solidworks knowledge while allowing me to refine and acquire new 3D modeling skills. This will provide useful experience with CAD and the overall design process that can be applied post-graduation.

Objectives:

- Practice accurate reverse engineering techniques
- Refine knowledge of use of calipers and other measurement tools
- Use various Solidworks commands to create parts, assemblies, drawings, and renderings
- Demonstrate learned skills from previous coursework

Deliverables:

- Documented reverse-engineered measurements
- Proper part files and drawings for components
- Sub-assemblies and one overall assembly
- Rendered 3D model of go-kart design
- Report detailing design process and final product

Background on Go-Kart

During the summer of 2019, I interned at Richard Childress Racing (RCR), a NASCAR team located in Welcome, NC. My friend Ryan was also interning in the same complex. My assigned work required that I interact with the fabricators in the RCR chassis shop. After becoming friends with some of the workers there, they offered to help Ryan and me fabricate a go-kart. I originally planned on modeling the go-kart using PTC Creo software, but I ended up not having enough time. The go-kart was made completely custom, the only constraint being that it needed to fit in the trunk of my SUV.

Process and Challenges

To model the components of the go-kart, I needed to use reverse engineering techniques to acquire accurate measurements and document the interfaces between different parts. The main tools I used were digital calipers and a measuring tape. I was unable to measure a portion of the components while they were still assembled, so I used wrenches and other tools to disassemble certain parts.

After gaining access to the main parts, I used rough sketches on engineering paper to record the measurements I was taking. Initially, I tried to take measurements of the main components, and ones that did not interface with each other. Doing this would allow me to model the remaining parts using the measurements obtained from the main components. The main components I examined first included the steering wheel, overall frame, spindles, floor plate, and the pedals.

Due to the go-kart being completely custom, no baseline dimensions were used when it was being fabricated. This led me to encounter numerous challenges during the modeling process. None of the dimensions I measured were common increments, and much less round numbers. Therefore, I decided to round my measurements on many of the components to the nearest 1/8”.

Using these dimensions, I modeled all of the parts of the go-kart, minus the engine, rear axle bearings, chain, clutch, and brake lines. For some of the more complex components, namely the brake calipers, I simplified the design and did not model the internal components in order to save time. Other than the above mentioned parts and the brake assembly, I aimed to model the components as accurately as possible using calipers.

For the hardware used on the go-kart, I downloaded Solidworks files from McMaster-Carr. McMaster-Carr provides downloadable models and drawings for nearly any kind of hardware. Using the models from McMaster-Carr saved time in the modeling process and allowed me to focus on the details of the other parts. Any of the nuts, bolts, bearings, threaded rods, or rod ends I used in my model can easily be found by searching the part number on McMaster-Carr’s

website. Not only are there models and drawings, but there is also ample information about the material properties of each piece of hardware.

Although using the hardware was beneficial, it still presented some challenges. The rod ends I used to connect the steering rods to the steering shaft were not flexible. Since I wanted the steering to be functional in the assembly file, I had to delete the body of the ball within the rod end and make an assembly with my own modeled ball. The same issue occurred for the rod ends along the steering shaft.

After modeling all of the different parts, I created small assemblies to be used in one overall assembly. This helped me keep the overall assembly clean and easy to edit. I made subassemblies out of the brake disc and mount, brake calipers and mount, brake master cylinder system, pedals and pedal rod, rear axle mounts, sprocket and mount, steering wheel and steering shaft, and the steering rods.

Finally, I created a rendering of the overall assembly. I experimented by changing the scenes, camera angle, depth of field, reflectivity, and brightness of the rendering. A low reflectivity setting and higher brightness made the go-kart appear similar in the rendering as it does in real life.

Part and Assembly List

Below is a list of the parts and assemblies I created, along with a part number. The individual component numbers start at #1001, and the assemblies start at #2001. This is a simplified version of a part system a company may use to keep track of their CAD files in a library style database.

Item Number	Component
1001	Brake Disc
1002	Brake Frame Mount
1003	Brake Disc Mount
1004	Brake Pedal
1005	Gas Pedal
1006	Engine Mount
1007	Floor Plate
1008	Floor Plate Tabs
1009	Frame
1010	Front Tire
1011	Front Wheel Hub
1012	Front Wheel Hub Connection
1013	Rear Tire
1014	Rear Wheel Hub
1015	Rear Wheel Hub Connection
1016	Half of Sprocket
1017	Spindle
1018	Pedal Rod

1019	Rear Axle
1020	Rear Axle Bearing Bracket
1021	Rear Axle bearing Mount
1022	Spindle Collar
1023	Sprocket Mount
1024	Steering Rod
1025	Steering Rod Standoff
1026	Steering Shaft
1027	Steering Wheel
1028	Brake Caliper
1029	Brake Lever
1030	Brake Mount
1031	Brake Master Cylinder
1032	Brake Piston
1033	Brake Pad
1034	Brake Lever Rod
1035	Brake Cylinder Seal
1036	Spindle Washer
1037	Steering Rod Heim
2001	Brake Disc Assembly
2002	Front Right Assembly
2003	Brake Cylinder Assembly
2004	Rear Axle Mount Assembly
2005	Brake Caliper Assembly
2006	Pedal Assembly
2007	Rear Brake Assembly
2008	Rear Hub Assembly
2009	Sprocket Assembly
2010	Steering Wheel Assembly
2011	Steering Rod Assembly
2012	Original Design Assembly

Book Exercises

All of the exercises completed came from "*Solidworks 2018 Intermediate Skills*" by Paul Tran. The following list shows the exercises I went through and what I learned.

Chapter 2: Sketching Handle

- This exercise used multiple circular patterns. I learned that you can select any circular edge in the model to use as the direction for the pattern, as long as it is concentric with the pattern circle. I also started to try using the view shortcuts that were shown in the beginning of the book. Previously, I would rely on moving my mouse or right-clicking on a plane and selecting the "Normal to" option. Using Ctrl + 1, 2, 3, 4, 5, 6, or 7 is a much more efficient way to utilize different views.

Chapter 2: Working with Sketch Pictures

- This exercise introduced me to sketching using pictures inserted into Solidworks. I learned how to scale a picture to a desired size and position it in different locations. One of the details this pointed out was that when sketching, the user can check to see if their sketch lines are connected by right clicking on a line and choosing the "Select Chain" option. This can come in handy when it might look like there is a closed sketch but it is not registering as a closed loop.

Chapter 2: Working with Sketch Pictures & Using the Spline tool

- This exercise involved tracing a more complex picture than the previous exercise. To trace this picture accurately, the spline tool had to be used. While playing around with the spline tool, I discovered that by using the handles connected to each point on the spline, the user can create infinite different arcs. In the past, I never utilized the spline tool because I was never able to get it to match the shape I was trying to sketch. However, spending time working with the spline handles has made me more comfortable with the tool and its capabilities.

Chapter 2: Re-use Geometry

- This exercise showed how a single sketch can be used to create multiple features. I have been in the habit of creating a new sketch for each feature I want to create, whether it be an extruded boss, extruded cut, revolve, etc. It can save large amounts of time if you don't have to create multiple sketches. This removes the process of having to convert different entities from prior sketches in order to create the desired new sketch.

Chapter 3: Creating Multibody Parts

- Multibody parts was a completely new topic to me. I was not aware that a single part file could contain separate bodies. This is a very important concept, and I was able to apply it while modeling components of the go-kart. I also gained practice using the Copy tool and creating exploded views.

Chapter 3: Combining Multibodies

- This exercise involved starting with one solid body, inserting another part, and mating them together in a single part file. Again, this topic was fascinating since I had never had more than one solid body in a part file. Another feature this exercise introduced was combining bodies. This was also a new topic for me. I learned that parts can be combined by adding to a single body, subtracting the overlapping material, or by keeping the common material that overlaps between multiple bodies.

Chapter 4: Working with Multibody Parts Creating Mates & Exploded Views

- This exercise introduced the possibilities with using numerous bodies in a Solidworks part file. However, I am unsure about the differences between creating an assembly file

with multiple parts compared to using multiple bodies. I believe being able to use multiple bodies can be useful when creating smaller assemblies because there will be a smaller number of files to manage.

Chapter 4: Working with Multibodies

- Once again, this exercise worked with multibody parts. It consisted of mating two parts. The combine feature was also used to eliminate certain features that were interfering between the two parts. The display view was altered as well, showing how a wireframe display can be useful in examining internal parts.

Chapter 5: Cylinder

- This exercise involved creating a routinely revolved part. In addition, threads were extruded onto the end the part. This was the first time I was introduced to using the "offset" command within the thread feature. Using this allows the thread to be continuous along the entire desired length. I also learned you can use the "trim with start/end face" selection in order to achieve the same effect.

Chapter 5: Revolved and Threads

- This exercise again provided practice with the "thread" feature. This time the threads were cut rather than extruded. I also learned how to manipulate a section view by cutting at the different planes that can be selected. Furthermore, I saw how a "Zonal" section view can be created by selecting two different cutting planes. Previously, I thought section views like this could only be achieved in drawings.

Chapter 6: Flex Bending Spanner

- Using the flex tool was entirely new to me. I had seen it in the "Tools" dropdown, but I had never used it. Changing the location of the planes and the "Triad" determined how the part was changed. It took me a while to understand how these changes impacted the end result. Additionally, the deform feature was used in this exercise. It was completely new to me as well. The deform feature allowed complex surfaces to take on a new profile from a sketch. I believe this could be useful in combination with imported images to create freeform bodies that have intricate curves.

Chapter 7: Sweep With Guide Curves Plastic Bottle

- This exercise illustrated how multiple guide curves can be used to create a curved object. Beyond that, I learned how to split a body on its surface, so that the outer surface itself can be edited. The surfaces were also knit back together and merged to create one continuous outer layer. This seems like a very powerful method for altering parts. However, I would need ample practice before I could start using this method of modeling regularly.

Chapter 7: Sweep with Solid Body

- This exercise showed how a sketch perpendicular to the sweeping profile can be used to create different end conditions of the feature. Using a cylinder in this example and cutting through a solid body resulted in a cut with curved ends. This would be helpful to know how to do when it comes to cutting through cylindrical features when the sweep needs to follow the curvature of the cylinder.

Chapter 7: Sweep vs. Loft

- This exercise detailed how using the loft command can sometimes be more appropriate than the sweep command. Using the sweep command without the proper guide curves can create unwanted twists in the profile along the path. I learned that by using the loft command, a centerline can be chosen instead of a path, resulting in a feature that is not twisted.

Chapter 8: Lofts and Boundaries Solar Boat

- This was the longest exercise I was able to complete. It was excellent practice for working with guide curves and lofts. I learned about using the different types of fillets, specifically the face fillet. The face fillet can be used when the fillet does not need to be uniform, and it creates a smooth, unbroken fillet. This exercise also consisted of splitting surfaces with a sketch like in a previous exercise. This time, however, the sketch was projected onto the surface to be split rather than sketched directly on the surface.

Chapter 8: Surface and Solid Modeling

- This exercise utilized the “Surface Offset” command. Once again, the surfaces were split so that they could be offset. This was a fantastic exercise for showing how additional features can be created by using already existing features.

Chapter 9: Surfaces and Patches Welding Fixture

- This exercise displayed how to use the “Trim Surface” and “Lofted Surface” commands. A surface can be trimmed using a sketch, and then the profile that is left can be used as a profile for a loft. This is useful in creating lofts that do not have flat end profiles. Additionally, I learned how the “Boundary Surface” command can be used to effectively stretch a surface.

Chapter 10: Configure Features Wooden Crate

- This exercise showed how a design table can be used to create multiple configurations of a part. I saw how certain bodies within the part can be suppressed/shown depending on the configuration. These configurations are controlled through the design table in Microsoft Excel. A useful feature is that multiple sizes of parts can also be set for configurations through the design table.

Conclusions and Benefits

Modeling various different components provided an opportunity to experiment with new modeling techniques. Ultimately, this was the overarching goal of the entire independent study. When modeling, there are numerous ways to generate a model. These methods vary in time and difficulty.

During my internship at Richard Childress Racing, it became apparent to me that when many people interact with the same parts and drawings, the way the part was created is very important. Parts need to be modeled intuitively, so that other people can understand what is going on within the feature tree. Overall, the more efficiently a part can be modeled, the better.

This semester allowed me to gain knowledge and practice with alternate techniques. For example, I learned how to model sheet metal parts and create drawings showing the flat pattern of a part. In addition, I utilized multibody techniques that I learned from the textbook exercises. If I had known about the features when I started modeling the go-kart components, I could have saved the time I spent going back and forth between parts to make sure their dimensions matched. I believe we briefly covered multibody parts in GC-350, although I did not entirely remember how to use them. Specifically, I used multibody parts to create models for the front and rear tires, to make them fit exactly on the complex profile of the wheel hubs. Multibody parts also proved useful for modeling the components of the brake master cylinder and the bar attaching the brake pedal to the lever operating the master cylinder.

Another technique I was able to use was creating coordinate systems. This also helped modeling the bar attaching the brake master cylinder to the brake pedal. I saw the benefit of creating new coordinate systems during my internship at Richard Childress Racing. All of their components had a coordinate system that allowed them to immediately be placed in the correct location in an assembly by mating the assembly origin and the part's coordinate system. During my internship, I used this feature in PTC Creo software, and I wanted to learn how to use it in Solidworks.

Lastly, I gained valuable experience working with assemblies and different kinds of mates. For some components, I inserted them directly into the overall assembly. Other components I put in subassemblies first, and then inserted the entire subassembly. I learned that a subassembly can be set as "rigid" or "flexible" in order to keep the animated motion. I practiced forming new subassemblies from selected components within the overall assembly as well. The large amount of hardware needed for the model helped me practice advanced and mechanical mates, notably the "Screw" mate. Consequentially, I ended up using the "Hole Wizard" tool to create threads on many of the components I modeled.

All in all, this study allowed me to gain proficiency in the techniques I had already established, while also introducing me to more efficient, higher level techniques. Modeling programs like

Solidworks have countless features and seemingly endless capabilities. Compared to the beginning of this semester, I believe I have widened my range of knowledge when it comes to Solidworks functions. Now, I can consider alternative ways for part and assembly creation in order to model in a more effective way.

Future

While completing this study did allow me to explore new features in Solidworks, there are several additional features and methods that I want to learn.

When creating the assemblies, I noticed that some parts did not fit correctly or needed to be edited in some other way. Since the assembly was quite large, I relied on opening the part files separately and making any necessary changes there. However, I saw that there was an "Open Part in Position" option. This opens the part in the same orientation and position as seen from the current view in the assembly file window. Using this could be beneficial when there are large, intricate parts incorporated into the assemblies.

In GC-350, we learned how to assign properties within a part file and link them to fields in a drawing template so that the properties would not need to be filled in manually. I did not do this for the parts I modeled, but I think it would have been very beneficial in terms of the amount of time it took for me to fill out each field in the individual drawings. In the future, I hope to implement the capabilities of the assigned part properties in order to create drawings more efficiently.

In the book exercises, I was introduced to surface modeling using Solidworks. Surface modeling was another method that I have experience with using PTC Creo. Surface modeling is the key to crating freeform parts and features. I plan on continuing to experiment with the surfaces package Solidworks has.

When modeling the frame for the go-kart, I mainly used the swept boss feature. While this method worked fine, I was unable to sweep circular profiles up to another feature. This may be something that is entirely possible, but I did not know how to accomplish it. My model ultimately became a combination of swept and extruded features. However, I believe either using surface and thickening them, or using the tubing tool in Solidworks would have been more efficient.

Eventually, I hope to learn how to use the analysis packages in Solidworks. Proficiency in analysis tools would be a great skill, and would also be useful in the industry. I plan on continuing my Solidworks education either by investing in online classes, video tutorials, or by buying a book similar to the one I worked out of this semester.